● BIBLIOMETRIC ANALYSIS · NEURORADIOLOGY ARTIFICIAL INTELLIGENCE

Artificial Intelligence Research in the *American Journal of Neuroradiology*

A decade of artificial intelligence in neuroradiology — from proof-of-concept to clinical decision support. Tracking publication volume, citation impact, the models and methods in use, clinical application areas, methodological rigor, and the people and places driving the field.

229 papers · 2017–2026 window · 5,456 total citations · 23.83 mean citations per paper · 9 subfields tracked

01 Executive Summary

Artificial intelligence research in the American Journal of Neuroradiology has matured from a nascent field into a dominant research area — exponential growth in output paired with significant, front-loaded citation impact.

229 Artificial intelligence papers (2017–2026)	**5,456** Total citations accrued	**55** Papers in 2025 — peak volume (9× 2017)	**2021** Peak impact year (235.7 age-adjusted citations)

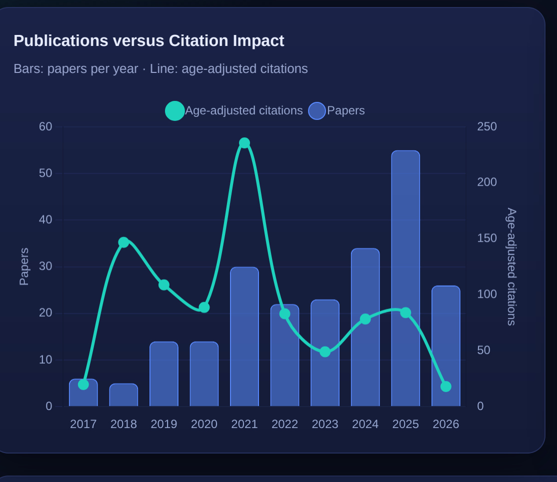
The core story: Publication volume surged from 6 papers (2017) to 55 (2025), while citation *impact* peaked earlier in 2021 — foundational work on clinical translation and multimodal integration gained traction before the volume-driven expansion. The field has shifted from early exploratory studies toward clinically focused applications: **artificial intelligence moving from proof-of-concept to clinical decision support.**

02 Year-by-Year Evolution

Volume and citation impact diverge: output keeps climbing while age-adjusted impact crested in 2021 and again shows resilience in 2025. Recent years naturally show fewer raw citations — papers have had too little time to accumulate them.

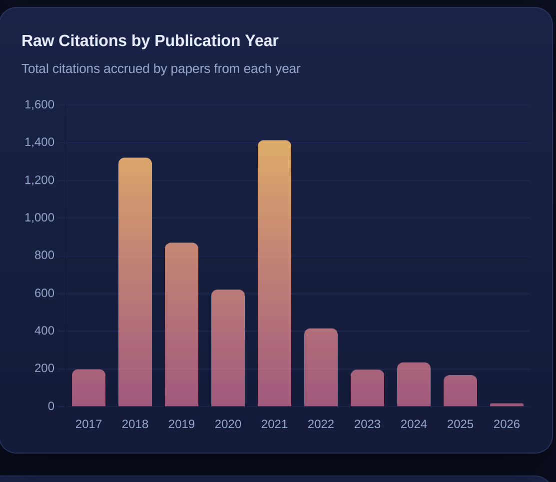
Publications versus Citation Impact

Bars: papers per year · Line: age-adjusted citations



Raw Citations by Publication Year

Total citations accrued by papers from each year



Detailed Yearly Breakdown

Counts, citations, and the leading subfields each year

YEAR	PAPERS	RAW CITATIONS	AGE-ADJUSTED CITATIONS	TOP SUBFIELDS (BY COUNT)
2017	6	198	19.8	Quantitative imaging (6) · Detection / classification (5) · Image segmentation (5)
2018	5	1,321	146.8	Detection / classification (5) · Image segmentation (4) · Quantitative imaging (4)
2019	14	870	108.8	Quantitative imaging (12) · Image segmentation (11) · Detection / classification (9)
2020	14	621	88.7	Quantitative imaging (12) · Image segmentation (10) · Detection / classification (9)
2021	30	1,414	235.7	Detection / classification (25) · Quantitative imaging (17) · Image segmentation (14)
2022	22	415	83.0	Quantitative imaging (13) · Image segmentation (11) · Prognosis / outcome prediction (11)
2023	23	196	49.0	Image segmentation (16) · Detection / classification (15) · Quantitative imaging (12)
2024	34	235	78.3	Detection / classification (27) · Image segmentation (19) · Quantitative imaging (17)
2025	55	168	84.0	Quantitative imaging (37) · Detection / classification (33) · Image segmentation (29)
2026	26	18	18.0	Detection / classification (19) · Quantitative imaging (19) · Image segmentation (13)

Top-cited paper per year — 2018: *Deep-learning convolutional neural networks classify genetic mutations in gliomas* (437) · 2019: *Deep learning detection of intracranial aneurysms in three-dimensional time-of-flight MR angiography* (160) · 2020: *Deep transfer learning and radiomics for high-grade glioma survival* (107) · 2021: *Artificial intelligence and acute stroke imaging* (228) · 2025: *Large language models in radiology: the road ahead* (21).

03 Subfield Growth & Decline

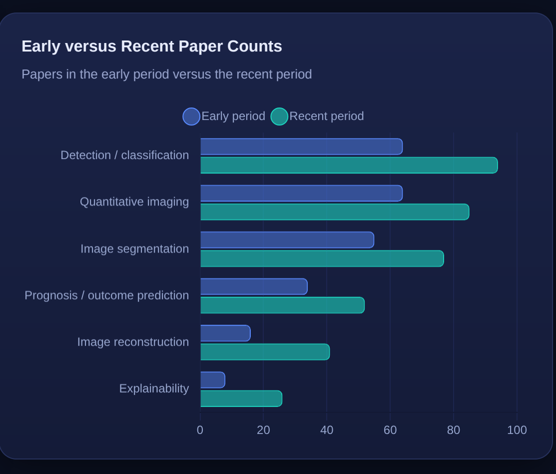
Prognosis / outcome prediction is the fastest-growing subfield, propelled by artificial intelligence's role in personalized treatment planning. A handful of fragmented, lowercase-duplicate labels nominally "declined" to zero — a taxonomy artifact rather than a real trend.

Fastest-Growing Subfields

Compound annual growth rate (percent)

Early versus Recent Paper Counts

Papers in the early period versus the recent period



Declining labels are duplicates, not real decline. "Machine learning (general)", "Machine learning", and a lowercase "multimodal AI" each fell from 1 to 0 papers. The main **Multimodal AI** subfield (12 papers) is actually growing. This labeling inconsistency is addressed in the recommendations.

04 Most-Cited Subfields

High-volume subfields dominate *total* citations; high-impact subfields (large language models / natural language processing, multimodal artificial intelligence) lead on *mean* citations per paper. The top three — detection / classification, quantitative imaging, image segmentation — make up roughly 80% of the corpus.

Total Citations by Subfield

Volume-driven scholarly footprint

Mean Citations per Paper

Impact density — niche subfields punch above their weight

SUBFIELD	PAPERS	TOTAL CITATIONS	MEAN CITATIONS
Detection / classification (computer-aided diagnosis)	158	4,554	28.82
Quantitative imaging	149	4,258	28.58
Image segmentation	132	3,936	29.82
Prognosis / outcome prediction	86	2,159	25.10
Large language models / natural language processing highest mean	34	1,268	37.29
Image reconstruction / acceleration	57	1,166	20.46
Explainability / trustworthy artificial intelligence	34	949	27.91
Generative / diffusion models	28	781	27.89
Multimodal artificial intelligence	12	424	35.33

05 Most-Cited Papers

The most influential work centers on high-impact clinical translation — convolutional neural networks and radiomics applied to stroke and tumor characterization. Four of the top five were published in the field's 2018 foundational moment.

01 **Deep-learning convolutional neural networks accurately classify genetic mutations in gliomas** — 437 CITATIONS
2018 · Integrates genetic classification, explainability and multimodal artificial intelligence — foundational for artificial-intelligence-driven oncology

02 **Radiomics in brain tumor: image assessment, quantitative feature descriptors, and machine-learning approaches** — 350 CITATIONS
2018 · Seminal radiomics review bridging imaging and machine learning for tumor characterization

03 **Deep learning in neuroradiology** — 274 CITATIONS
2018 · Comprehensive overview — established artificial intelligence as a core neuroradiology discipline

04 **Hybrid three-dimensional / two-dimensional convolutional neural network for hemorrhage evaluation on head computed tomography** — 244 CITATIONS
2018 · Early clinical translation for acute stroke; high adoption in emergency settings

05 **Artificial intelligence and acute stroke imaging** — 228 CITATIONS
2021 · Real-world clinical deployment — reflects field maturation

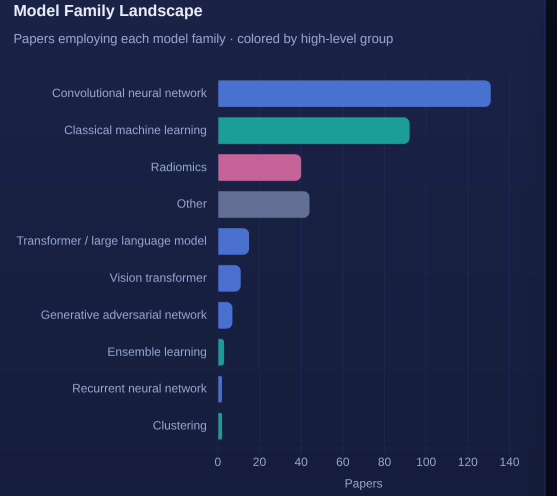
Common thread: all integrate clinical translation, use high-impact methods (convolutional neural networks, radiomics), and target high-need applications (stroke, tumor classification).

06 Models & Methods

What kind of artificial intelligence does the field actually use? At the highest level the corpus splits into deep learning, classical machine learning, radiomics (feature-based), and other. Underneath, specific architectures and algorithms differ sharply in citation impact. Papers may employ more than one approach, so counts overlap rather than partition the corpus.

Model Family Landscape

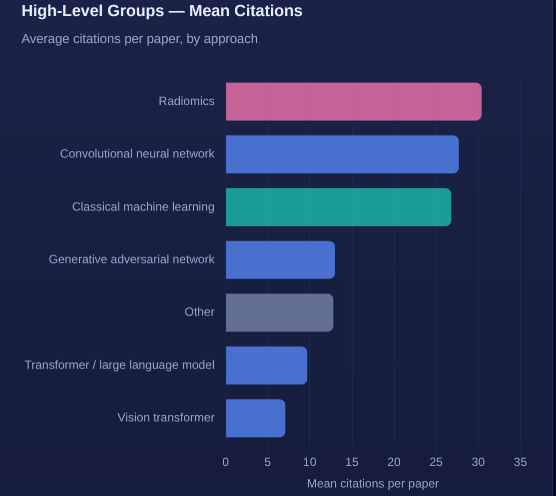
Papers employing each model family · colored by high-level group



Deep learning (blue) dominates by volume — convolutional neural networks alone appear in 131 papers. **Classical machine learning** (teal) remains substantial (92 papers), while **radiomics** (pink) is the highest-impact family by mean citations (30.4).

High-Level Groups — Mean Citations

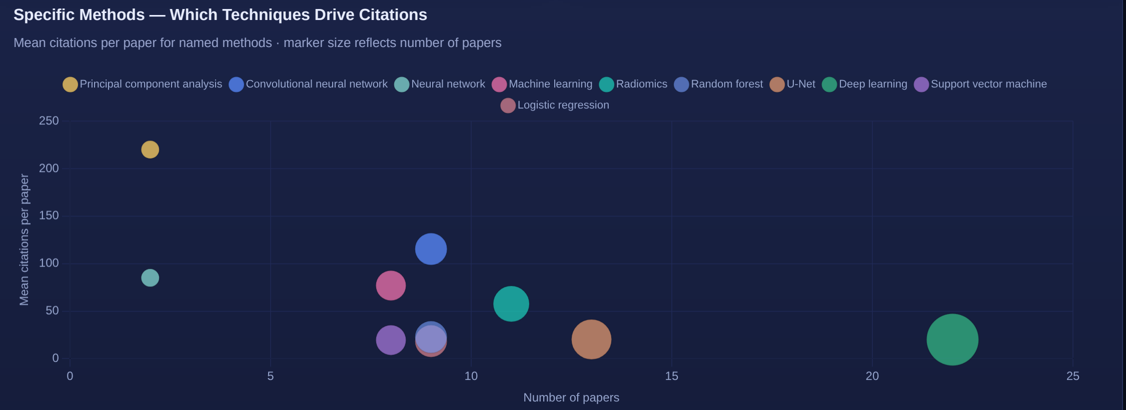
Average citations per paper, by approach



Newer deep-learning variants (transformers / large language models, vision transformers, generative adversarial networks) still carry **low mean citations** — partly because most are recent and have had little time to accumulate citations.

Specific Methods — Which Techniques Drive Citations

Mean citations per paper for named methods · marker size reflects number of papers



Several extreme means rest on very few papers — **principal component analysis** (220.0) spans just 2 papers and **convolutional neural network** (115.3) just 9. Treat low-count means as suggestive, not definitive. The broad, robust workhorses are deep learning (22 papers), U-Net (13), radiomics (11), random forest (9), logistic regression (9) and support vector machine (8).

Model Families — Detail

MODEL FAMILY	PAPERS	TOTAL	MEAN
Convolutional neural network deep learning	131	3,628	27.7
Classical machine learning	92	2,462	26.8
Radiomics highest mean	40	1,214	30.4
Other	44	563	12.8
Transformer / large language model deep learning	15	146	9.7
Generative adversarial network deep learning	7	91	13.0
Vision transformer deep learning	11	78	7.1
Ensemble learning	3	55	18.3
Recurrent neural network deep learning	2	17	8.5
Clustering	2	3	1.5

Specific Methods — Detail

METHOD	PAPERS	TOTAL	MEAN
Principal component analysis low n	2	440	220.0
Convolutional neural network	9	1,038	115.3
Neural network low n	2	170	85.0
Machine learning	8	615	76.9
Radiomics	11	634	57.6
Random forest	9	204	22.7
U-Net	13	262	20.2
Deep learning	22	439	20.0
Support vector machine	8	156	19.5
Logistic regression	9	166	18.4

07 Clinical Application Areas

Mapping the field by clinical target — which diseases and body systems the highest-impact work addresses. Neuro-oncology and cerebrovascular disease anchor the most-cited cluster, with pediatric / fetal imaging and broad cross-cutting reviews rounding out the landscape.

Neuro-oncology
GLIOMA · BRAIN TUMOR · GENETIC CLASSIFICATION
- Convolutional neural networks classify genetic mutations in gliomas · 2018 — 437
- Radiomics in brain tumor: features & machine learning · 2018 — 350
- Deep transfer learning & radiomics for high-grade glioma survival · 2020 — 107

Cerebrovascular & Stroke
ACUTE STROKE · HEMORRHAGE · CLINICAL DEPLOYMENT
- Hybrid convolutional neural network for hemorrhage on head computed tomography · 2018 — 244
- Artificial intelligence and acute stroke imaging · 2021 — 228
- Stroke augmented intelligence & communications platform · 2023 — 32

Vascular & Aneurysm
INTRACRANIAL ANEURYSM · ANGIOGRAPHY
- Deep-learning detection of intracranial aneurysms in time-of-flight MR angiography · 2019 — 160

Pediatric & Fetal
FETAL BRAIN · PEDIATRIC BRAIN TUMORS
- Automated three-dimensional fetal brain segmentation with deep learning · 2022 — 52
- nnU-Net and DeepMedic for autosegmentation of pediatric brain tumors · 2024 — 28

Cross-Cutting & Methods
FIELD REVIEWS · LARGE LANGUAGE MODELS · ACCELERATION
- Deep learning in neuroradiology (review) · 2018 — 274
- Large language models in radiology: the road ahead · 2025 — 21
- Deep-learning–based acceleration in MRI · 2026 — 5

How to read this: the source data does not include a structured per-disease tag, so this is a qualitative map built from the clinical targets of the most-cited and annual-leading papers — **not** a corpus-wide frequency count. A dedicated disease-tagging pass over all 229 papers would be needed to quantify prevalence and citation share per disease.

08 Methodological Rigor & Open Science

Gradual maturation: external validation and multi-center data rose from 0% (2017) to peaks around 2021–2024. But **open code remains near zero** and **open data is entirely absent** across the decade — the field's clearest unmet standard.

Open-Science Indicators Over Time

Share of papers per year meeting each rigor criterion (percent)

External validation and multi-center adoption climbed steadily; code availability first appears only in 2025 (3.6%); open data never exceeds 0%. The 2025 code uptick signals nascent — but fragile — open-science adoption.

09 Influential Authors & Institutions

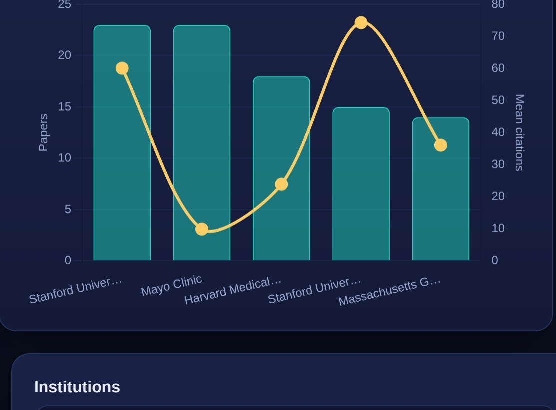
A few authors drive outsized influence through concentrated, high-impact work. Leadership is dominated by United States institutions and split between high-volume contributors and high-impact-per-paper centers.

Top Authors — Output versus Impact

Bars: total citations · Line: mean citations per paper

Top Institutions — Volume versus Mean Impact

Bars: papers · Line: mean citations per paper



Authors

AUTHOR	PAPERS	CITATIONS	MEAN	ACTIVE
Zaharchuk G.	18	1,085	60.3	2018–2025
Filippi C.G. top mean	6	968	161.3	2018–2024
Wintermark M.	12	929	77.4	2018–2025
Yeom K.W.	7	636	90.9	2018–2024
Chang P.D.	8	537	67.1	2018–2026

Institutions

INSTITUTION	PAPERS	CITATIONS	MEAN
Stanford University School of Medicine	23	1,382	60.1
Mayo Clinic high volume, low impact	23	228	9.9
Harvard Medical School	18	430	23.9
Stanford University top mean	15	1,114	74.3
Massachusetts General Hospital	14	505	36.1

10 Notable Gaps & Recommendations

Four structural gaps emerge — and each maps to a concrete recommendation for the American Journal of Neuroradiology and the wider neuroradiology artificial intelligence community.

Open-science deficit
Open code availability remains near zero (3.6% in 2025); open data is entirely absent across the decade.
→ Prioritize open-science mandates in artificial intelligence submissions.

Subfield fragmentation
Duplicate and case-variant labels (for example, "detection / classification" appearing in several casings) distort growth analysis.
→ Standardize subfield taxonomy in future bibliometrics.

Emerging-trend blind spot
No reliable data on emerging keywords, methods, or benchmarks could be extracted from the corpus.
→ Track keyword bursts and benchmark adoption longitudinally.

Geographic concentration
United States institutions dominate (14 of 15 top institutions); no non-United-States leaders appear in the rankings.
→ Encourage global collaboration in artificial intelligence neuroradiology research.

Source: `ajnr_ai_trends_report_thinking.md` and `ajnr_ai_trends_report_thinking_facts.json` · Bibliometric analysis of artificial intelligence research in the American Journal of Neuroradiology, 2017–2026 (229 papers, 5,456 citations). Charts rendered client-side with Chart.js. Recent-year citation counts are immature and not directly comparable to earlier years. The Clinical Application Areas section is a qualitative map derived from the most-cited papers, not a structured per-disease count.